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Concept and design: Arnett, Harstad, O'Connell, Brewster, Barbaresi, Doan.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: Arnett, O'Connell, Barbaresi, Doan.
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Country-Level Improvements in Nurturing Care and Child Development

In 2018, the World Health Organization, UNICEF, and the World Bank launched the Nurturing Care Framework, a political "roadmap for action" to guide countries' investments in young children's responsive caregiving, early learning, safety and security, health, and nutrition.[1] Although nurturing care has shown robust cross-sectional associations with individual child outcomes,[2] evidence quantifying changes in and benefits of societal investments in care remains limited. We documented changes in provision of nurturing care over the past 10 years in 31 low- and middle-income countries (LMICs) and examined whether these changes were associated with improvements in countries' child development outcomes.

Supplemental content

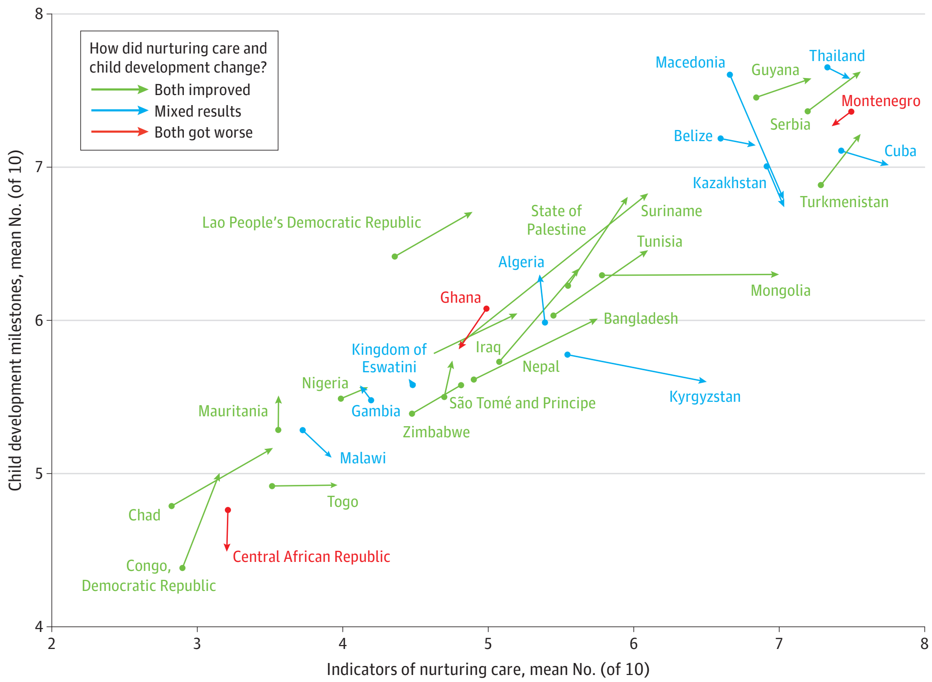
Methods | We used data from UNICEF's Multiple Indicator Cluster Surveys (MICSs), a series of multinational, population-representative household surveys.[3] We included all LMICs with at least 2 nationally representative MICSs completed between 2010 and 2019 that included data on nurturing care and child development for children aged 3 and 4 years. We used multiple imputation to address missingness in study variables at the child level. This cross-sectional study was considered not human participants research according to US federal guidelines and therefore did not require institutional review board review. We followed the STROBE reporting guideline for observational studies.

Following prior work,[4] we summed each child's access to 10 indicators of nurturing care (eg, participation in early childhood care and education; access to adequate water, sanitation, and hygiene; and absence of physical punishment). We measured each child's development using the sum of 10 caregiver-reported physical, literacy-numeracy, social-emotional, and approaches to learning milestones from the Early Childhood Development Index (ECDI).[5] We used MICS-provided sampling weights to calculate country-level means in nurturing care and child development within each imputed survey.

We examined country-level changes in nurturing care and child development descriptively. We also regressed country-level changes in child development from time 1 to time 2 on concurrent country-level changes in nurturing care, controlling for changes in countries' Human Development Index and the number of years between surveys. This analysis incorporated uncertainty due to imputation and weighted observations by total within-country sample size (ie, number of children at time 1 plus time 2). All analyses were completed in Stata 17.0 and in R version 4.3.1 using RStudio and Tidyverse.

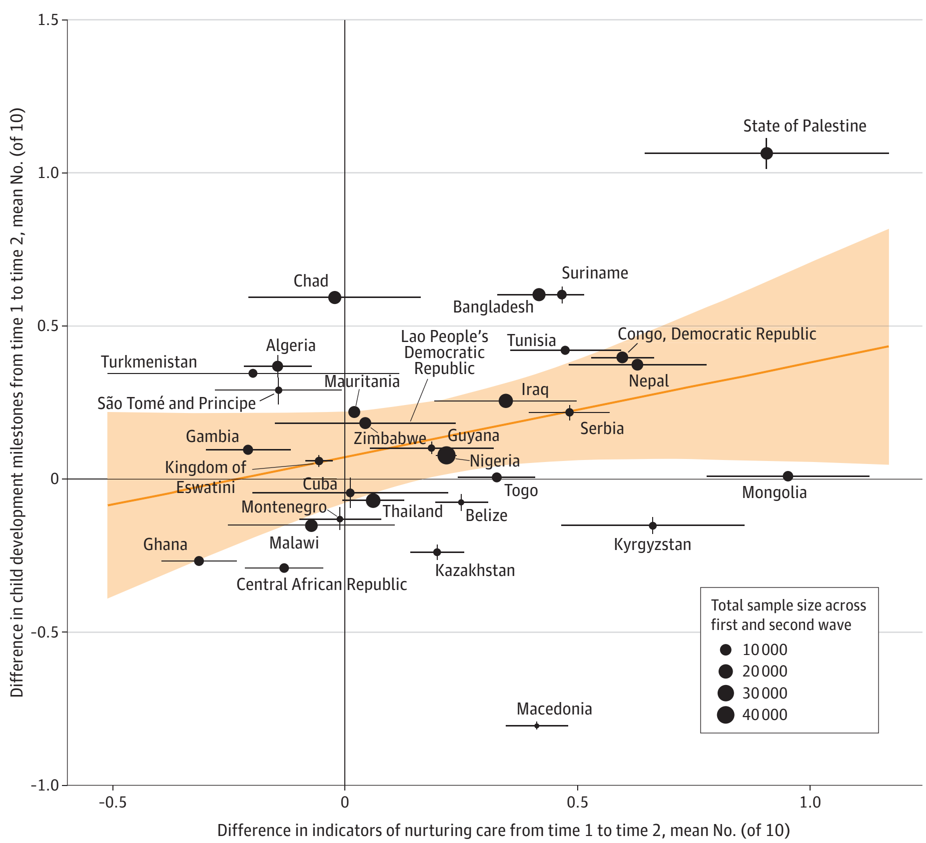
Results | The final sample included 220 759 children aged 3 and 4 years (49.3% female) living in 31 LMICs. Country-level nurturing care and child development are visualized over time in **Figure 1**. Of 31 countries, 21 improved nurturing care during the mean 6.06 (range, 4-9) years between surveys. During this time, countries improved from providing a mean of 5.53 (range,

Figure 1. Changes in Nurturing Care and Early Childhood Development in 31 Low- and Middle-Income Countries



Country means were pooled across 10 imputations to incorporate uncertainty due to child-level missingness.

Figure 2. Associations Between Changes in Nurturing Care and Early Childhood Development



Differences were calculated on the basis of changes in country means that were pooled across 10 imputations to incorporate uncertainty due to child-level missingness. Bars represent 95% credible intervals; orange line, the line of best fit; and shaded area, SE of the line of best fit.

2.96-7.90) to 5.74 (range, 3.20-8.16) of 10 nurturing care indicators. Changes in nurturing care ranged from −0.06 (Turkmenistan) to 0.19 (Mongolia) indicators per year.

Of 31 countries, 21 also improved in child development. Between surveys, countries improved from achieving a mean of 6.05 (range, 4.39-7.64) to 6.19 (range, 4.47-7.61) of 10 ECDI milestones. Changes in child development ranged from −0.12 (Macedonia) to 0.12 (State of Palestine) milestones per year.

Associations between country-level changes in nurturing care and child development are visualized in **Figure 2**. Regression analyses showed that each additional indicator of nurturing care provided by countries was associated with an additional 0.36 (SE = 0.15; *P* = .03) developmental milestones.

Discussion | Since 2010, both nurturing care and child development improved modestly within 31 sampled LMICs. Improvements in countries' provision of nurturing care were associated with gains in children's early development, even when accounting for time trends and improvements in countries' general human development. Although our analysis is limited in causality, representativeness, and breadth and depth of measurement, these findings are the first, to our knowledge, to document potential societal (rather than individual) benefits of nurturing care. Results suggest that additional resources, services, and policies targeting both families and broader communities are needed to support countries in achieving global policy benchmarks related to child well-being, including Sustainable Development Goal 4.2.[6] Further investments in measuring nurturing care and enabling environments are also needed to better monitor countries' progress and to identify novel targets of intervention.

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Concept and design: All authors.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: All authors.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: All authors.
Administrative, technical, or material support: McCoy.
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Comparison of Methods to Assess Adolescent Gender Identity in the ABCD Study

While reliable and valid methods to assess gender identity exist for adults, their utility and feasibility are less well understood for adolescents. For adults, the 2-step method, asking sex assigned at birth and current gender identity, is recommended for clinical settings and population-based studies[1] as it accurately captures gender diverse (GD) identities that differ from those typically associated with assigned sex (eg, transgender, nonbinary).[2-4]

Supplemental content

Methods | This cross-sectional study was approved by the University of California, San Diego central institutional review board, and parent and child written informed consent or assent was obtained. The study followed the STROBE reporting guideline.

The Adolescent Brain Cognitive Development Study is among the first nationally to use the 2-step method with adolescents. At the 3-year follow-up visit (age, 12-13 years), youths were asked to report their assigned sex on their birth certificate (male, female, don't know [DK], refuse to answer) and current gender identity (boy, girl, another gender [eg, nonbinary], I don't understand the question [IDU], refuse to answer). A separate measure asked, "Are you transgender? Yes, maybe, no, IDU, refuse to answer."[5] Data from the 5.0 release, in the National Institute of Mental Health Data Archive, include the full cohort at the 3-year visit and approximately half at the 4-year visit. Responses to both 2-step items (10 273 youths at 3 years and 4720 at 4 years) were analyzed for feasibility (rates of refusal, DK, IDU) and to compare methods of identifying GD adolescents. Data analysis was performed from June 21 to July 14, 2023, using SPSS, version 29.0 (IBM Corporation). A 2-sided $P < .05$ was considered significant by χ^2 test.